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Version 1.1

10 February 2019



Introduction

This is a copy of my original web page started in September 2002 and describes my experience with the RACAL RA17 receiver. It comprises my initial impressions of the radio as well as a brief description of the front panel controls. Next is a brief explanation of the NATO stock number system as well as a summary of the restoration work I carried out on my radio. Finally, there is a brief description of the Wadley Loop principle.

This particular set seems to have had several owners and I've been told it originally started its life at [Radio Australia](#). Along the way it has suffered its share of distress and modifications that required a little work to resolve.

What's so good about them?

For me, the key to their appeal is three-fold.

The first lies in their application of the Wadley Loop. This effectively means they have incredible frequency stability and tuning accuracy for a receiver of its time. The overall frequency stability of this set is governed by one single crystal, something unheard of at that time. The RA17 was the first mass produced set to use the Wadley Loop.

The tuning mechanism is another attraction. The design of the dial allows for easy setting to within 1 KHz. This was amazing for a time where frequency counters were not normal on receivers.

Finally, their construction. There's something about a radio that's built like a tank that really appeals.

These sets saw extensive use in the armed forces in many parts of the world. They were also used by government agencies, embassies, coast watch stations, shortwave broadcasters and frequency regulatory agencies around the world. Many, like mine, remained in service until the 1980's and will continue to provide great service into the new century for their lucky owners. Not bad for a set that ceased production in 1962.

The first thing to remember about sets like this is that they are probably vastly different from anything you have ever seen or used before. If you're any younger than forty it is highly unlikely that your radio experience would include something like this. Your experience with radios possibly amounts to tuning in the shortwave or ham bands on a dual conversions radio (tube or semi) by tuning up and down a tiny little dial and selecting the appropriate mode (am, cw, ssb). Or, it could be as simple as punching in a frequency on a digital receiver or spinning the dial

until the digital display shows the frequency of interest. Not that there's anything wrong with that. That's progress.

These sets need to be operated with care and finesse to get the best out of them. If you want to flick the set on and hear music and not fiddle about then this may not be the best set for you. If on the other hand you love valve gear and love the technology of it all and tweaking to get the best out of a radio then this is the one for you.

Using the RACAL RA17.

To familiarize myself with the set I followed the operations guide in the service manual. There are a number of major differences between this set and a many of today's more modern sets. Spending a little time thinking about the set and it's functions will pay dividends later on.

Kilocycle tuning.

The KHz tuning dial is flywheel weighted and it takes almost one complete revolution of the dial to travel 100 KHz. The KHz tuning scale of the RA17 is made from something like 35mm film and is calibrated in major increments of 10KHz along with minor increments of 5KHz and 1 KHz. The 50KHz and 100 KHz points and the 0 KHz and 1000 KHz are also clearly indicated. A mechanical cursor adjustment is built into the dial escutcheon to permit accurate calibration. At a glance it is a simple matter to identify your receive frequency to the nearest KHz within any 1 MHz band.

Six foot tuning dial?

The manual quotes an effective dial length of 145 feet. What does this mean? The distance between any two 100KHz points on the dial scale is six inches. If you had a continuous dial scale with the resolution of 6 inches per 100KHz, it would have to be 145 feet long if you were to tune from 1MHz to 30MHz. Consider that with some general purpose SW receivers you might go from 2MHz to 6MHz when you tune the dial six inches and you can start to see the appeal of a set like this.

What does the RED scale indicate?

The red scale is used in conjunction with the RA37 and RA137 LF Adapters. This allows the RA17 to tune from 10KHz to 980KHz. Due to the frequency conversion scheme used by these adapters the set tunes backwards when tuning below 1MHz. For this reason, you will find the red scale is a mirror image of the normal tuning scale.

But the scale goes to zero anyway?

True, but according to the manual performance below 1MHz down to 500KHz is possible with "slight degradation of performance". I've tested mine down to around 200KHz listening for non directional beacons (NDB) and found that it works OK but there is a definite drop off in performance at frequencies below 500KHz. I think the reason the manual quotes degraded performance is because there is no facility on the pre-selector for frequencies below 500KHz.

Tuning linearity.

Whilst the film scale is linear it's clear that no analog tuning system could ever hope to match the linearity of a scale that is 5 feet long. RACAL included a 100KHz calibrator to allow for slight tuning variations. Basically you tune the KHz dial to the 100 KHz point closest to your frequency of interest and zero beat to the internally generated 100KHz calibration signal. Once you have zero beat, manually adjust the KHz cursor on the escutcheon to coincide with the 100 KHz point on the film scale. The kilocycle dial is now calibrated. In this way you minimize tuning errors. I calibrated my receiver at the 500 KHz point on the dial and found that the maximum error at 1000KHz was -1.5 KHz and at 0KHz it was +1 KHz. This means that for most of the tuning range on my receiver the reading error will remain under 1000 cycles. Not bad considering the set was last calibrated in 1986.

Megacycle tuning.

The megacycle dial is simply marked in one MHz increments from 0 to 29 MHz. Select the Mhz band of interest and that's it, away you go. The tuning system on the Racal is amazing. This would have to be the easiest tuning system I have ever used. This is all due to the Wadley loop design. One dial for Mhz and one for KHz. You can't get much simpler than that.

Dial Lock.

Because the tuning is so smooth and easy to turn, RACAL included a dial lock to protect against accidental bumping. It's a simple friction lock that pushes a rubber stopper against the dial mechanism for both the KHz and MHz dials.

Best tuning system ever?

In later years a number of manufacturers incorporated the Wadley Loop into their design. Amongst the more famous sets available to the public were the Barlow/Wadley XCR-30 and the Yaesu FRG-7. In my opinion the "one dial for Mhz and one dial for KHz" system is clearly the most elegant and simple arrangement ever conceived for analog receivers.

Pre-selector and Attenuator.

The RA17 has a five band pre-selector including a wide-band setting. The pre-selector is not often seen in modern receivers and is used to manually tune the input of the receiver to the desired receive frequency. Frequencies outside the range of the pre-selector are severely attenuated. The wideband setting is used to by-pass the pre-selector when searching over a large range of frequencies. Situated just below the pre-selector is the attenuator. This is a five step attenuator allowing 0 - 40db of attenuation in 10db steps. Careful use of the pre-selector and attenuator will help avoid cross-modulation where strong unwanted signals are present near the frequency of interest.

Signal Meter.

The signal meter can be switched to shows either AF output or RF level. When set to AF, the audio output level for an external 600Ω output is monitored. This

output was designed to be connected to a landline in the case where the receiver may have been installed at a remote listening post. The monitored AF level can be controlled via a screwdriver adjustment hidden under the MHz dial. When switched to RF, the meter shows the current passing through the detector diode. In this way the meter can be used as a rudimentary albeit un-calibrated signal strength indicator. RA17's destined for the American market also included an S-meter setting that was calibrated in 4db per s-point increments.

The BFO.

The B.F.O. uses a reduction drive to allow accurate setting of the B.F.O. frequency over a range of $\pm 8\text{kc/s}$. The B.F.O. is exactly tuned to a central point on the I.F. amplifier response when B.F.O. NOTE KC/S control is set to zero-beat with the check bfo function on the calibrator.

Having standardized the B.F.O. frequency, the frequency of an incoming signal may be accurately measured by setting the KILOCYCLES control to a zero-beat position. The B.F.O. should be detuned in order to produce an acceptable note for c.w. and ssb reception.

A.V.C. - Automatic Volume Control.

AVC is designed to keep the audio output level regardless of the signal strength of the received signal. It does this by adjusting the receiver gain depending on the strength of the incoming signal. In the RA17, the BFO is injected into the detector, this has the effect of reducing the receiver gain (sensitivity) because the AVC reacts to the BFO signal as it would to any other received signal. Trying to resolve strong SSB signal under these conditions can be very difficult if not impossible. CW reception is not much better. To get around this problem the designers made it possible to disable the AVC and control the receiver gain manually. With judicious use of the RF/IF gain control and AF level the results for both SSB and CW can be very satisfying. The AVC has a switchable long and short time constant. The choice of time constants depends largely on conditions.

Using manual gain.

Using a receiver with manual gain control is strange at first but once you've fiddled with it for a while it becomes much easier. The first thing to do is back the RF/IF gain right off and open the AF level right up. You may hear the audio change and become harsh as you do this. Adjust the AF level to just below the point where the harshness sets in. Your AF level is now set correctly. The audio output will now be controlled solely by the RF/IF gain control. Now bring the RF/IF gain up until you can hear signals at a reasonable level. With my set I found the best results were achieved with a detector current of 80 - 100 μA as shown on the meter. Tuning around the bands is a bit of an experience. It's quiet compared to tuning with AVC. Remember that AVC controls receiver gain to try and maintain a constant audio output. With manual gain you are in charge and must ensure that the gain control is at its optimum setting. Have the gain set too low and you might miss a faint signal. Have the gain set too high and you may

overload the receiver. Using manual gain requires patience and a light touch but once mastered becomes second nature.

Limiter.

The limiter on this set is particularly effective with impulse noise. Electric fence or gas starter noises are almost completely eliminated. The limiter works by clipping any signal exceeding modulation peaks of 30%. Using the limiter when listening to good quality signals will result in deterioration of the recovered audio. As it says in the manual "It does not introduce noticeable distortion below a 30% modulation level". Maybe it should read "It will introduce noticeable distortion for signals with greater than 30% modulation level". Still, if you are listening to faint signals this might just make the difference between making the copy or losing it.

Bandwidth Selection.

Six bandwidths are provided as follows:-

13 kc/s, 6.5 kc/s, 3 kc/s and 1.2 kc/s (L-C filter)

300 c/s and 100 c/s (crystal filter)

The two crystal filters determining the narrowest bandwidths are adjusted to ensure that their centre frequencies are within 50 c/s of each other, it follows that any filter width can be selected without retuning the receiver. The 13 kc/s and 6.5 kc/s settings are great for listening to A.M. signals.

The Speaker.

If there is one thing that disappoints about this set - it has to be the speaker. There is a 2.5 inch speaker on the front panel that could be used to monitor the receiver output. I doubt anyone could bear to listen to that speaker for more than half an hour. The response, understandably, is awful. Clearly in a monitoring situation the line out or headphone output are much preferred. I imagine that the designers felt the speaker would only be used during the initial installation of the set and as such did not need to be anything special. Good idea back then, too bad for us. An external speaker does wonders for this set.

On the air.

On air tests show a great receiver that is quite capable of matching performance with today's modern receiver.

Frequency stability and accuracy is excellent. On initial testing I came across a QSO on 40M and I was able to determine the frequency of the stations as 7085.2 on the RACAL. A quick check on my Kenwood R-2000 and Yaesu FRG-8800 showed the same result. There is no discernible drift over an extended period of listening.

Sensitivity is great and the RACAL can hear everything that my R-2000 and FRG-8800 can hear.

In the selectivity department I think the RACAL is ahead. With careful use of the pre-selector and manual gain I feel the RACAL just edges out the two more modern receivers.

Recovered audio is not as good as the R-2000 but I feel the Kenwood audio is very hard to beat. The audio from the RACAL is certainly of a high enough standard to allow listening over an extended period - as long as you use a decent speaker!

Overall a wonderful receiver that is just as good if not better than many modern receivers.

Repair and restoration

This particular set seems to have had several owners and I've been told it originally started its life at [Radio Australia](#). The person I got it from had kept it wrapped in plastic whilst in storage and had not turned it on since 1997.

I'm not sure about the origins of this radio. There is a label inside the radio that states it was calibrated for Radio Australia in 1986. However there is also a NATO identification plate attached to the front of the set.



Ex-services Racal equipment is fitted with a front panel identification plate comprising of the N.S.N (Nato stock number), description, serial number and the date code. This is in addition to the Racal serial number plate fitted on the rear of the equipment. The military plate gives either a letter code or a figure code which indicates when the equipment was manufactured. This date code may appear in the form 'UD' or '7315'.

A Jan	M 1955
B Feb	N 1956
C Mar	O 1957
D Apr	P 1958
E May	Q 1959
F June	R 1960
G July	S 1961
H Aug	T 1962
I Sept	U 1963
J Oct	V 1964
K Nov	W 1965
L Dec	X 1966
	Y 1967
	Z 1968

After the end of 1968, the date code changed to a numerical presentation giving the year and week of manufacture. Thus '7315' indicates the equipment was manufactured during week 15 of 1973. However, it is possible in some instances that the label may show the actual year of manufacture i.e. 1957.

The ex-services front panel identification plate is usually black with the relevant info stamped onto it. Non-services equipment front panel identification plate is usually all blue with just the equipment description included on it.

This would indicate that it was used in a military capacity for at least some part of it's life. The date code indicates that it was made some time in February 1962.

Along the way it has suffered it's share of distress and modifications that required a little work to resolve.

- Missing rear identification plate. A cross check of the NATO stock number on the front of the set reveals that it's a RA17L.
- Missing glass from tuning dial. I didn't notice for a while that the glass was missing until I touched the dial scale by accident. A new piece of Perspex glued in place soon fixed that.
- I felt the MHz dial was a little off and found the best position for the mc dial was 'about' 500 KHz above the desired mc band. A tune through the MHz ranges showed the amount error was consistent. I adjusted the scale of the mc dial so the mc points on the scale coincided with the centre of the pass band. In hindsight, it may be that the alignment was OK and the mc dial was calibrated to indicate the two points between which the radio would tune.
- The Burndept aerial socket was replaced with a BNC socket. For some reason the person that changed the socket saw fit to cut a piece out of the back panel. This piece unfortunately included the brass binding post normally situated below the aerial socket.
- One of the headphone jacks on the front panel had been replaced. While it worked ok, a large heavy duty shake proof washer holding it in place looked a bit tacky so I replaced both jacks with some new high quality shielded types.
- The HT fuse holder had been bypassed. Sometime in the past the fuse had blown or was broken and the end of a fuse was still jammed in the body of the fuse holder. A similar size fuse holder was obtained to fit in place. Take care if you need to do something similar as space is tight and too much movement of the solid core wires could result in breaks.
- Mains cable hard wired and tie-wrapped into place. This is probably the worst thing that has happened to the set. The original Plessey socket has long gone and although the wiring was well insulated, using tie-wraps to anchor a mains cable is dangerous and just plain poor workmanship.
- Missing alignment tool and Allen keys. Not a big deal but if you come across one and wonder what those bits of metal are on the top cover, they are meant to hold Allen keys. There is a clip on one of the filter cans inside the set that should also hold an alignment tool.
- General damage to the top cover. Someone dropped something sharp and heavy on the top of the case. It took a bit of work with a mallet to sort

out the damage. A little work on the Allen key holder got them back in working order too.

- Very heavy staining from cigarette smoke. At first I thought the outer panels were gold anodized aluminium. A closer inspection revealed the truth. All covers were removed from the set and lots of work with a scrubbing brush and household cleaner and the garden hose brought them back to their original color. Whilst the inside of the set was similarly affected it was not possible to use the same treatment to clean in there. Instead I used alcohol swabs to carefully clean all surfaces that I could reach. The underside of the set was also affected by the smoke but I thought it best to leave it be.
- The knobs and switches were removed from the front panel and the panel was unbolted from the chassis. Be careful not to lose the nuts holding the panel to the chassis. A gentle wash with a soft cloth and soapy water brought the panel back to almost new condition. Take care when cleaning the front panel as the lettering can easily be removed by solvent style cleaners. All the knobs were scrubbed with a toothbrush in warm soapy water then rinsed in clean water and left to dry. A little Liquid-Paper® in the dots and pointers on the knobs brought them up like new.

Power-up.

Power-up was uneventful with all tubes lighting and supplies settling to the prescribed values. About 25 seconds after power-up noise rushed from the speaker. After 8 years of silence the RA17 lives again. Note that with the exception of the mains cable and fuse holder and phone socket replacement there was no other electrical or alignment work involved in the restoration of this set.

Receiver Alignment.

The following note in the service manual makes it quite clear that alignment is generally not required.

WARNING! The receiver will, under normal conditions, remain in alignment over an extremely long period time, consequently ALL POSSIBILITY OF OTHER CAUSES OF LOW SENSITIVITY SHOULD BE ELIMINATED BEFORE RE-ALIGNMENT IS CONSIDERED, and should then only be undertaken by order of the Engineer responsible for the maintenance of the equipment.

Elsewhere in the manual I found this,

With the exception of V5, replacement of valves will not affect receiver alignment.

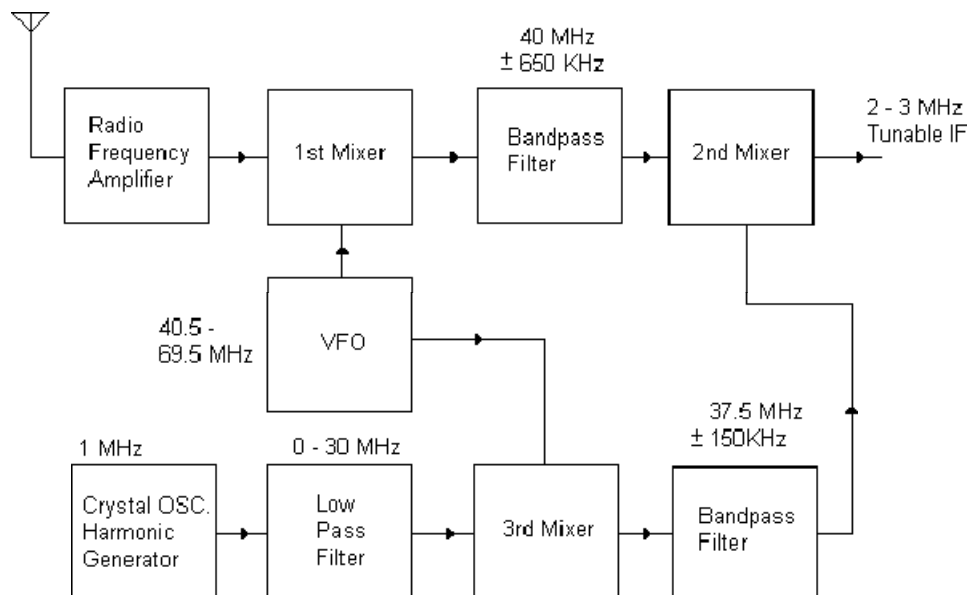
Clearly, these sets are built to last and remain reliable well after they are taken out of service. If you must fiddle or tweak do it with the utmost of care and make sure you are doing it for the right reason.

The Wadley Loop

The following is an excerpt from a detailed description of a solid state communications receiver based on the Wadley loop principle. It was published as a series of articles in Electronics Australia from January 1971 to May 1971. The author of the construction article was the late Ian Pogson. Part one of the series is presented here because I think it explains the Wadley loop very well. You will note a strong similarity to the RACAL block diagram. Applying this description to the diagram in the RACAL manual should not prove too difficult.

"When the writer was in the United Kingdom around 1959, a receiver made by the "Racal" organisation was brought to his attention at an exhibition. In common with many other receivers, this one is based on the familiar super heterodyne principle. However the actual system employed is a development of the basic superhet principle, and is so ingenious that full marks for originality must be awarded to the originator, Dr Wadley, of South Africa. The principle is now known as that of the "Wadley Loop". The Racal receiver covers continuously from 1MHz (it is possible to go lower) to 30MHz, in steps of 1MHz, with remarkable frequency stability. Yet it uses only one crystal. In effect, the Wadley Loop principle produces the equivalent of 29 crystal-locked converters, which may be accommodated in little more space than is normally occupied by one conventional crystal-locked converter.

The system functions in the following manner, which may be understood by reference to the block diagram below.



Incoming signals, in the range 1-30 MHz are selected by the RF amplifier, which is switched and tuned in the same way as in a conventional full coverage receiver. The RF amplifier is designed so as to have as uniform gain as possible

over the complete range to be covered. However, while the stage is normally peaked on the desired signal, adjacent signals will be passed to an extent, depending upon its selectivity.

After amplification by the RF stage, the signals are passed to the first mixer. The output from the mixer feeds into a band-pass filter having a flat top centred on 40MHz. The bandwidth of the filter is $\pm 650\text{KHz}$, the reason for which should become apparent later. The signal which passes through this filter may be considered to be the first IF of the receiver.

The other signal fed to the first mixer is the output from the VFO, The frequency of the VFO is tuneable over the range which extends from 1 MHz above the lower limit of the band-pass filter, to 30MHz above this frequency: from 40.5MHz to 69.5MHz. It is the VFO frequency which therefore, by heterodyne action, determines which particular 1 MHz band of input RF signals is "translated" into the (40MHz $\pm 500\text{KHz}$) IF range accepted by the band-pass filter. The VFO tuning control thus becomes the 1MHz selector of the receiver.

After passing through the band-pass filter, the IF signals derived from the chosen 1MHz band is fed to one input of the second mixer. The function of this mixer will be examined in a moment. At this stage it should be noted that the operation is very similar to that of a normal super heterodyne receiver, with some change in the IF and its bandwidth.

In another part of the receiver, however, signals are produced by a 1MHz crystal oscillator and harmonic generator, and fed through a low-pass filter with cut-off at 32MHz. The resultant series of harmonics are fed to a third mixer.

The output of the VFO is also fed to this mixer and heterodynes with the harmonics from the crystal oscillator. The heterodyne, in which interest centres, is that which happens to fall, in the course of VFO setting, within the pass-band of a second substantially flat topped band-pass filter on 37.5MHz $\pm 150\text{KHz}$.

It is the amplified signal on 37.5MHz which becomes the injection for the second mixer. This frequency, when heterodyned against the IF signals passed through the 40MHz band-pass filter, results in an output on 2.5 MHz $\pm 500\text{KHz}$, or in other words 3-2MHz.

This becomes the tuneable IF, which is passed on to the rest of the receiving set-up. It is in the following tuneable IF section that the individually desired signals are selected, the preceding circuitry having served only to change the incoming signals from their original frequency to within the 2-3MHz range.

And here an important point should be made in relation to the frequency stability of the system.

Any frequency drift in the VFO will cause a change in the frequency of signals passing through the 40MHz band-pass filter. Simultaneously, there will be a deviation from the nominal 37.5MHz signal which is injected into the second mixer. These differences exactly cancel at the second mixer and consequently the tuneable IF is not changed. In fact, no adverse effects occur provided that the VFO resultants stay within the pass band of the two filters.

In effect, the only independent frequency determining device in the front end is the 1MHz crystal oscillator. The net result is a system with a very high degree of stability. In a complete receiver, the overall frequency stability is virtually determined by the local oscillator used in conjunction with the tuneable IF. And with careful design, it is possible to obtain good stability here as well.

Some practical examples will now be given to illustrate the foregoing description of operation. Consider that it is desired to receive a signal on exactly 21MHz. The RF stage will be set to pass this frequency. The VFO is set to 61.5MHz and this will beat with the signal on 21MHz to give a difference of 40.5MHz, which passes through to the second mixer.

The 61.5MHz from the VFO will also beat in the third mixer, with the 24th harmonic from the crystal to give a difference of 37.5MHz. This is passed through the 37.5MHz amplifier-filter, to the second mixer. The 40.5MHz and 37.5MHz produce in the output of the second mixer, a difference of 3MHz. This is acceptable to the tuneable IF at one end of its range.

Now, taking a second look, assume that the same 21MHz signal is to be received, but this time, with the VFO set to 60.5MHz. Obviously enough, this time a difference frequency of 39.5MHz will be produced, which is still passed by the filter to the second mixer.

The 60.5MHz from the VFO will now beat in the third mixer, with the 23rd harmonic from the crystal to give a difference of 37.5MHz and so to the second mixer. The 39.5MHz and 37.5MHz produce, in the output of the second mixer, a difference of 2MHz. This is also acceptable to the tuneable IF, at the other end of its range.

From these two examples, it is evident that each integral megacycle point is tuneable at both ends of the scale, which end being determined by the setting of the VFO. The exceptions to this are those frequencies at the start and finish of the overall coverage, normally 1MHz and 30MHz.

The dial of the VFO, which is the "MHz" selector, is not calibrated according to its actual frequency but rather to that band of frequencies which the receiver actually tunes. For example, 41.5 is calibrated as 1MHz, when 1-2MHz is covered; 42.5 as 2MHz, when 2-3MHz is covered, and so on.

To illustrate the effect of slight mis-setting, or drift of the VFO, the first example will be referred to again. The 21MHz signal is to be received, but the VFO is either set to or has drifted to 61.6MHz. This gives a difference frequency of 40.6MHz from the first mixer. Now, although the band-pass filter is only required to pass $40\text{MHz} \pm 500\text{ KHz}$, it may be recalled that it has been made slightly wider, ie, $\pm 650\text{KHz}$. This provision allows the 40.6MHz signal to pass to the second mixer.

The 61.6MHz from the VFO beats in the third mixer, with the 24th harmonic from the crystal oscillator to produce 37.6MHz. It has been pointed out already that the amplifier-filter is on $37.5\text{MHz} \pm 150\text{ KHz}$. Therefore, the 37.6MHz signal will still be passed to the second mixer. The 40.6MHz and 37.6MHz signals are

heterodyned together to give a difference of 3MHz at the output of the second mixer. This is exactly the same as the resultant frequency at this point as given in the first example, demonstrating the way in which drift in the VFO is automatically cancelled.

Should the VFO drift by an amount which runs the signals concerned down the skirts of the two filters, the frequency of the received signal will not be changed, but the signal strength will drop due to attenuation in the filters. This must be avoided for obvious reasons. In practice, the VFO can be made adequately stable and no such trouble is experienced. From the foregoing details, this system may be seen to offer many desirable features, which include: -

1. Continuous frequency coverage from 1-30 MHz.
2. Constant tuning rate.
3. Reasonably good band spread, ie, 1 MHz per sweep of the dial.
4. High order of frequency stability.
5. Excellent repeatability due to electronic rather than mechanical band switching.

Many other features could be added to the list. However, another point is worthy of note. Given a high grade dial on the tuneable IF, which has only to cover 1000 KHz and which effectively: "interpolates" between the settings of the MHz dial, the frequency of the received signal may be read off quite accurately. This is a very good feature where it is necessary to pre-set the receiver in readiness for an expected signal. As may be expected, a price must be paid for all the advantages which this system provides. Efficient low-pass and band-pass filters are required, in order to obtain correct operation. In addition, considerable care must be taken to shield various sections from each other, to prevent the harmonics from the 1MHz crystal from being injected and received as incoming signals. Power supply leads must also be filtered, where necessary, for the same reason.

Be this as it may, it is a small price to pay for the facilities which are available. No amount of writing can illustrate the delightful ease with which tuning can be carried out. A few minutes at the controls are all that is necessary to convince the operator."